

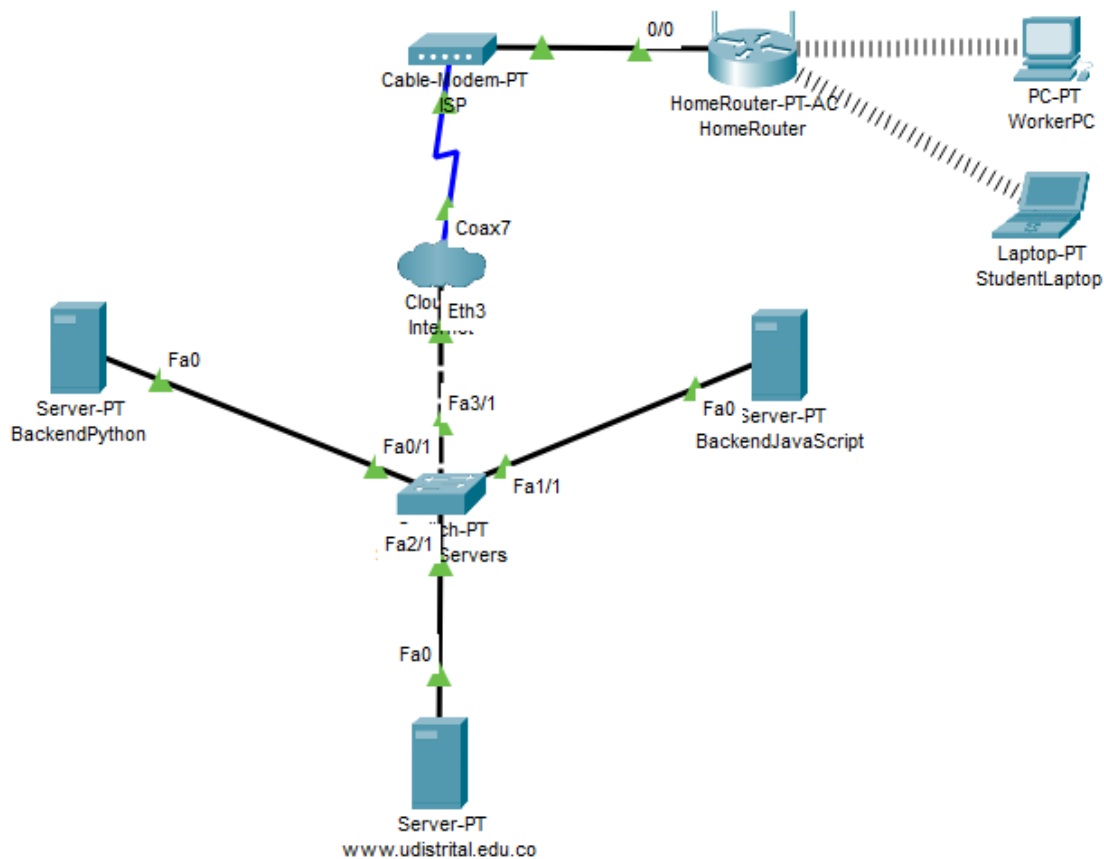
DISTRITAL FRANCISCO JOSE DE CALDAS COLLEGE

Workshop No. 2 – Computer Networking

Sockets and Services

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Design (with ports):



Technical decisions

This design apart from the technical decisions made on the last workshop add the following ones: the “Cloud” with a coaxial cable works on the side of the ISP, because it is cheap and works for the coaxial port on the ISP (Coaxial 7 in Cloud) so we connected a PT-Switch (Switch Servers) as connection of cable on Ethernet 3 of the cloud to connect both networks; we choose a Switch because it is a device that connects and routes well multiple devices with some security on it, and

also a VLAN can be made later with the help of this device, everyone of the servers (Backend Python, Backend Javascript, Frontend www.udistrital.edu.co) connected on Fast Ethernet ports of the switch through copper straight through cable and RJ45 connectors so it emulates the job of some datacenter facilities.

Test results

This test was run in the “Student Laptop”, we added a HTTP service on the server with the following HTML file (Index.html):

```
<!DOCTYPE html>
<html>
<head>
  <title>Workshop 4 - Networks</title>
  <meta charset="UTF-8">
  <meta name="description" content="This is a simple example to explore OSI layers">
  <link rel="stylesheet" type="text/css" href="styles.css">
</head>
<body>
  <h1>Workshop 4 - Networks</h1>
  <h2>OSI Layers</h2>
  <p>Click on the buttons to explore services on the network</p>
  <div>
    <button onclick="callPython()">Python Message</button>
    <button onclick="callJavaScript()">JavaScript Message</button>
  </div>
  <div id="result"></div>

  <footer>
    Created by Eng. Carlos Sierra / Eng. Omar Neira
  </footer>

  <script src="functions.js"></script>
</body>
</html>
```

With the styles.css:

```
body {
background-color: #1a1a1a;
color: #e0e0e0;
font-family: sans-serif;
margin: 0;
```

```
display: flex;
flex-direction: column;
min-height: 100vh;
}

h1, h2 {
color: #c0392b;
text-align: center;
margin-bottom: 10px;
}

p {
line-height: 1.6;
margin-bottom: 20px;
text-align: center;
}

div {
text-align: center;
margin-bottom: 20px;
}

button {
background-color: #8e4444;
color: #e0e0e0;
border: none;
padding: 10px 20px;
margin: 0 10px;
border-radius: 5px;
cursor: pointer;
transition: background-color 0.3s ease;
}

button:hover {
background-color: #c0392b;
}

#result {
background-color: #2c3e50;
padding: 20px;
border-radius: 5px;
margin: 0 auto;
max-width: 600px;
color: #e0e0e0;
text-align: center;
```

```

}

footer {
margin-top: auto;
background-color: #1a1a1a;
color: #95a5a6;
text-align: center;
padding: 10px 0;
border-top: 1px solid #34495e;
}

body {
box-shadow: 0px 0px 10px rgba(0, 0, 0, 0.2);
}

```

And the functions.js:

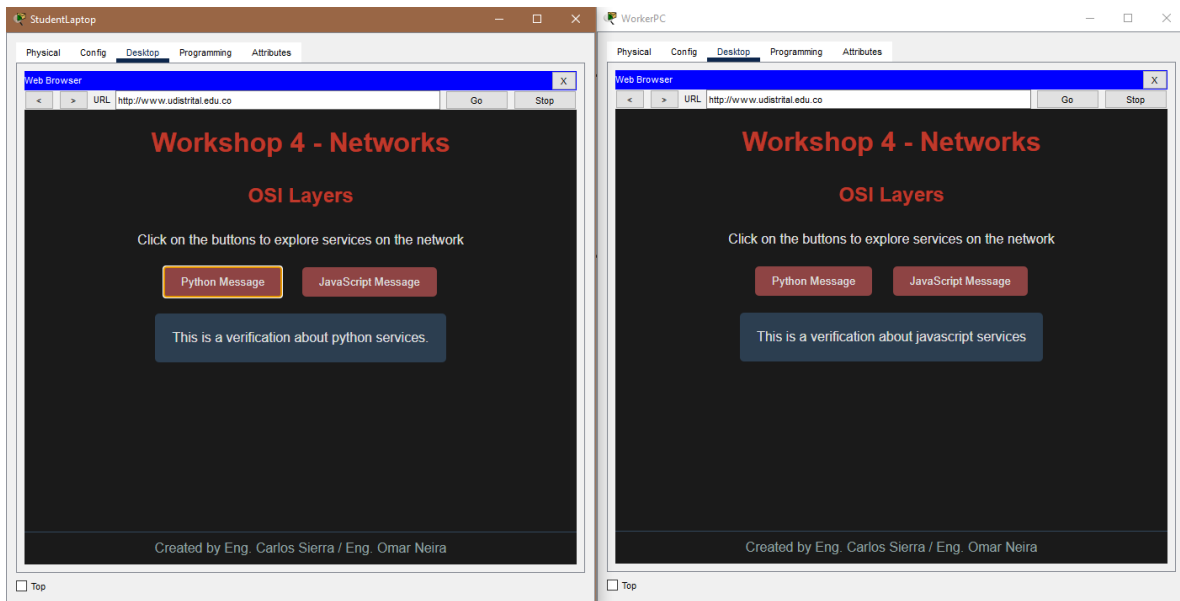
```

function callPython() {
let resultDiv = document.getElementById("result");
resultDiv.innerHTML = "Calling server on Python..."
fetch('http://193.168.100.201/healthcheck').then(response =>
response.text()).then(data => {
resultDiv.innerText = data;
}).catch(error => {
resultDiv.innerHTML = "¡Error!: "+error;
});
}

function callJavaScript() {
let resultDiv = document.getElementById("result");
resultDiv.innerHTML = "Calling server on JS..."
fetch("http://193.168.100.202/healthcheck").then(response =>
response.text()).then(data => {
resultDiv.innerHTML = data;
}).catch(error => {
resultDiv.innerHTML = "¡Error!: "+error;
});
}

```

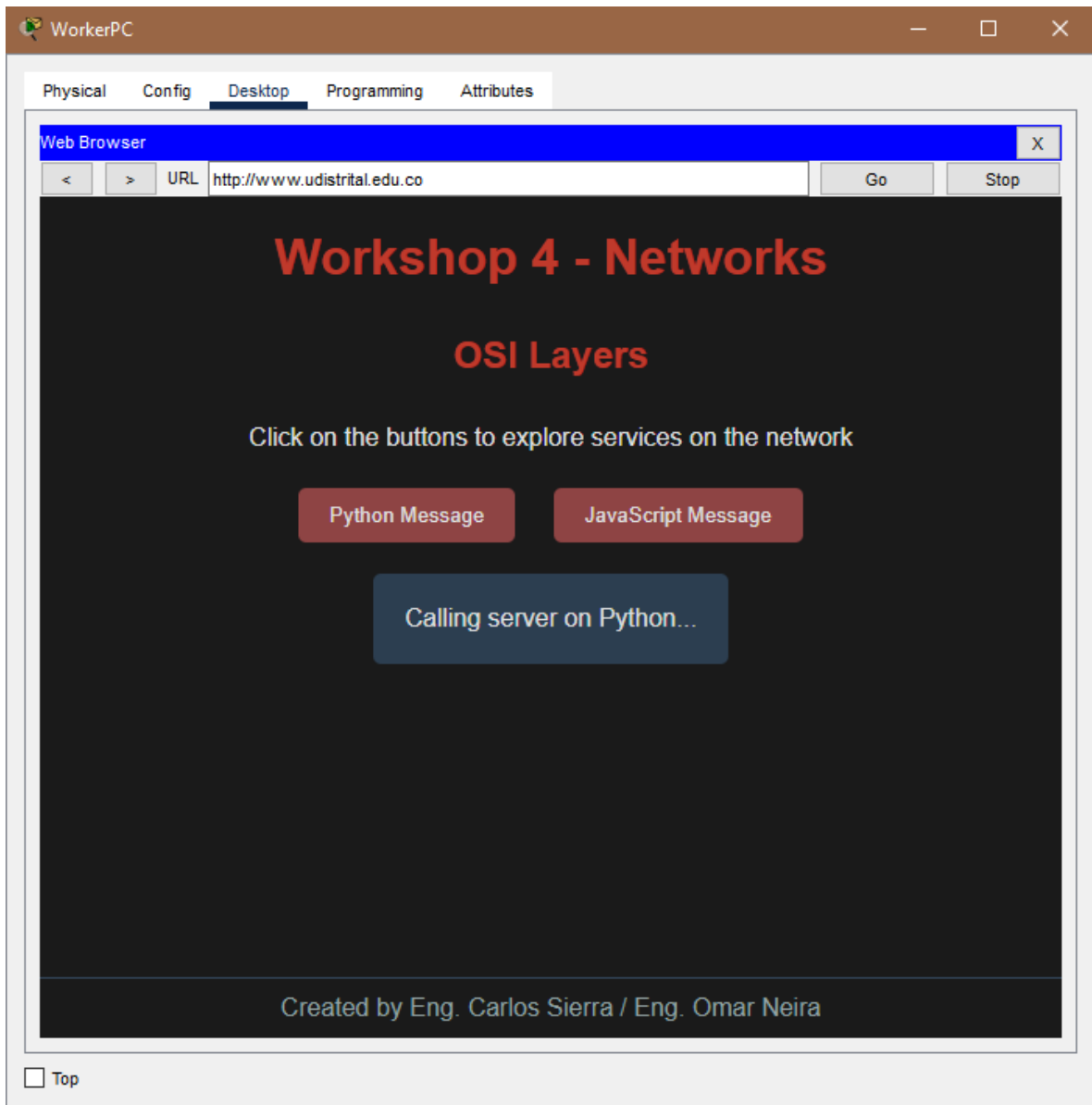
It was designed to be lightweight, then typing the URL “www.udistrital.edu.co” on the internet browser makes this output:



In both you can see what happens when you press the corresponding buttons to the backend to Python and Javascript.

OSI Layers

As soon one of the buttons is pressed, I put a text to wait for the response on the servers.



The OSI Layers looks like this when the PC starts making the request:

PDU Information at Device: WorkerPC

OSI Model

Outbound PDU Details

At Device: WorkerPC
Source: WorkerPC
Destination: 193.168.100.201

In Layers

Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1

Out Layers

Layer 7:
Layer6
Layer5
Layer 4: TCP Src Port: 1118, Dst Port: 80
Layer 3: IP Header Src. IP: 192.168.0.102, Dest. IP: 193.168.100.201
Layer 2: Wireless
Layer 1: Port(s): Wireless0

1. The HTTP client makes a connection to the server.

Challenge Me

<< Previous Layer

Next Layer >>

Here we can see that the transport layer (4) recognized the port of the server that we configured on Backend Python which is the port 80, on network layer (3) it recognized the IP of the device who's making the request and the destination IP, on the Data Link layer (2) says that the connection is wireless because the Worker PC is connected through the Wi-Fi of the Home Modem, and the physical layer (1) says that it is transported through the wireless port.

PDU Information at Device: HomeRouter

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: HomeRouter
Source: WorkerPC
Destination: 193.168.100.201

In Layers

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 192.168.0.102, Dest. IP: 193.168.100.201

Layer 2: Wireless

Layer 1: Port Wireless 1

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 193.168.100.1, Dest. IP: 193.168.100.201

Layer 2: Ethernet II Header 0090.2186.D301 >> 0004.9A27.1EB6

Layer 1: Port(s): Internet

1. Wireless 1 receives the frame.

Challenge Me

<< Previous Layer

Next Layer >>

In this next step we are on the Home Modem and it adds MACs addresses on the data link layer.

PDU Information at Device: Internet

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: Internet
Source: WorkerPC
Destination: 193.168.100.201

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer 1: Port Coaxial7

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer 1: Port(s): Ethernet3

1. Coaxial7 receives the frame.

Challenge Me

<< Previous Layer

Next Layer >>

The information is received to the ISP through the physical layer Coaxial cable.

PDU Information at Device: SwitchServers

[OSI Model](#)
Inbound PDU Details
Outbound PDU Details

At Device: SwitchServers
Source: WorkerPC
Destination: 193.168.100.201

In Layers

Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 0090.2186.D301 >> 0004.9A27.1EB6
Layer 1: Port FastEthernet3/1

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 0090.2186.D301 >> 0004.9A27.1EB6
Layer 1: Port(s): FastEthernet0/1

1. FastEthernet3/1 receives the frame.

Challenge Me
<< Previous Layer
Next Layer >>

Next the switch receives the information in the first and second layer and recognized which servers is the one that is required.

PDU Information at Device: BackendPython
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OSI Model
Inbound PDU Details
Outbound PDU Details

At Device: BackendPython
 Source: WorkerPC
 Destination: 193.168.100.201

In Layers

Layer7
Layer6
Layer5
Layer 4: TCP Src Port: 1118, Dst Port: 80
Layer 3: IP Header Src. IP: 193.168.100.1, Dest. IP: 193.168.100.201
Layer 2: Ethernet II Header 0090.2186.D301 >> 0004.9A27.1EB6
Layer 1: Port FastEthernet0

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Out Layers

Layer7
Layer6
Layer5
Layer 4: TCP Src Port: 80, Dst Port: 1118
Layer 3: IP Header Src. IP: 193.168.100.201, Dest. IP: 193.168.100.1
Layer 2: Ethernet II Header 0004.9A27.1EB6 >> 0090.2186.D301
Layer 1: Port(s): FastEthernet0

1. FastEthernet0 receives the frame.

Challenge Me

<< Previous Layer

Next Layer >>

The Switch sends the information to the Backend python server as it was sent in the first step, it decoded the Transport layer information such as the ports (80), IPs on the networks layer, MACs on the data link layer and through which port on the physical layer (FastEthernet0 of the BackendPython server).

PDU Information at Device: SwitchServers

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: SwitchServers
Source: WorkerPC
Destination: 193.168.100.201

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer 2: Ethernet II Header
0004.9A27.1EB6 >> 0090.2186.D301

Layer 1: Port FastEthernet0/1

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer 2: Ethernet II Header
0004.9A27.1EB6 >> 0090.2186.D301

Layer 1: Port(s): FastEthernet3/1

1. FastEthernet0/1 receives the frame.

Challenge Me

<< Previous Layer

Next Layer >>

The servers sends back the information required to the switch.

PDU Information at Device: Internet

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: Internet

Source: WorkerPC

Destination: 193.168.100.201

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer 1: Port Ethernet3

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer 1: Port(s): Coaxial7

1. Ethernet3 receives the frame.

Challenge Me

<< Previous Layer

Next Layer >>

The information is sent to Cloud through the physical layer.

PDU Information at Device: ISP

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: ISP

Source: WorkerPC

Destination: 193.168.100.201

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer 1: Port Port 0

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer 1: Port(s): Port 1

1. Port 0 receives the frame.

Challenge Me

<< Previous Layer

Next Layer >>

Now the information is received by the ISP also through the physical layer.

PDU Information at Device: HomeRouter

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: HomeRouter
Source: WorkerPC
Destination: 193.168.100.201

In Layers

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 193.168.100.201, Dest. IP: 193.168.100.1

Layer 2: Ethernet II Header 0004.9A27.1EB6 >> 0090.2186.D301

Layer 1: Port Internet

Out Layers

Layer7

Layer6

Layer5

Layer4

Layer 3: IP Header Src. IP: 193.168.100.201, Dest. IP: 192.168.0.102

Layer 2: Wireless

Layer 1: Port(s): Wireless 1

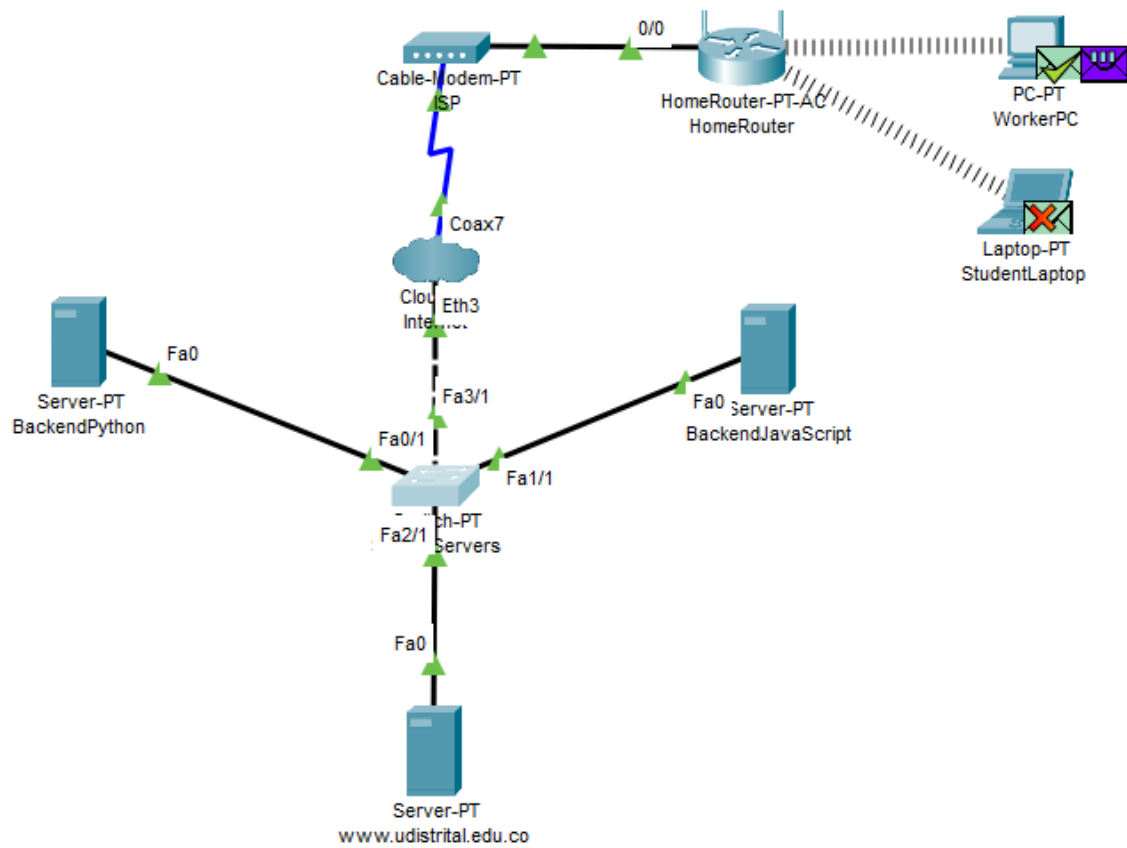
1. Internet receives the frame.

Challenge Me

<< Previous Layer

Next Layer >>

The ISP sends the information to the HomeRouter and it starts decoding it, so it can recognize the destination MAC and IP.



We can see that the request was made by the WorkerPC and not the StudentLaptop so it checks it as correct device.

PDU Information at Device: WorkerPC

OSI Model

Outbound PDU Details

At Device: WorkerPC
Source: WorkerPC
Destination: HTTP CLIENT

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer1

Out Layers

Layer 7: HTTP

Layer6

Layer5

Layer 4: TCP Src Port: 1118, Dst Port: 80

Layer 3: IP Header Src. IP: 192.168.0.102, Dest. IP: 193.168.100.201

Layer 2: Wireless

Layer 1: Port(s):

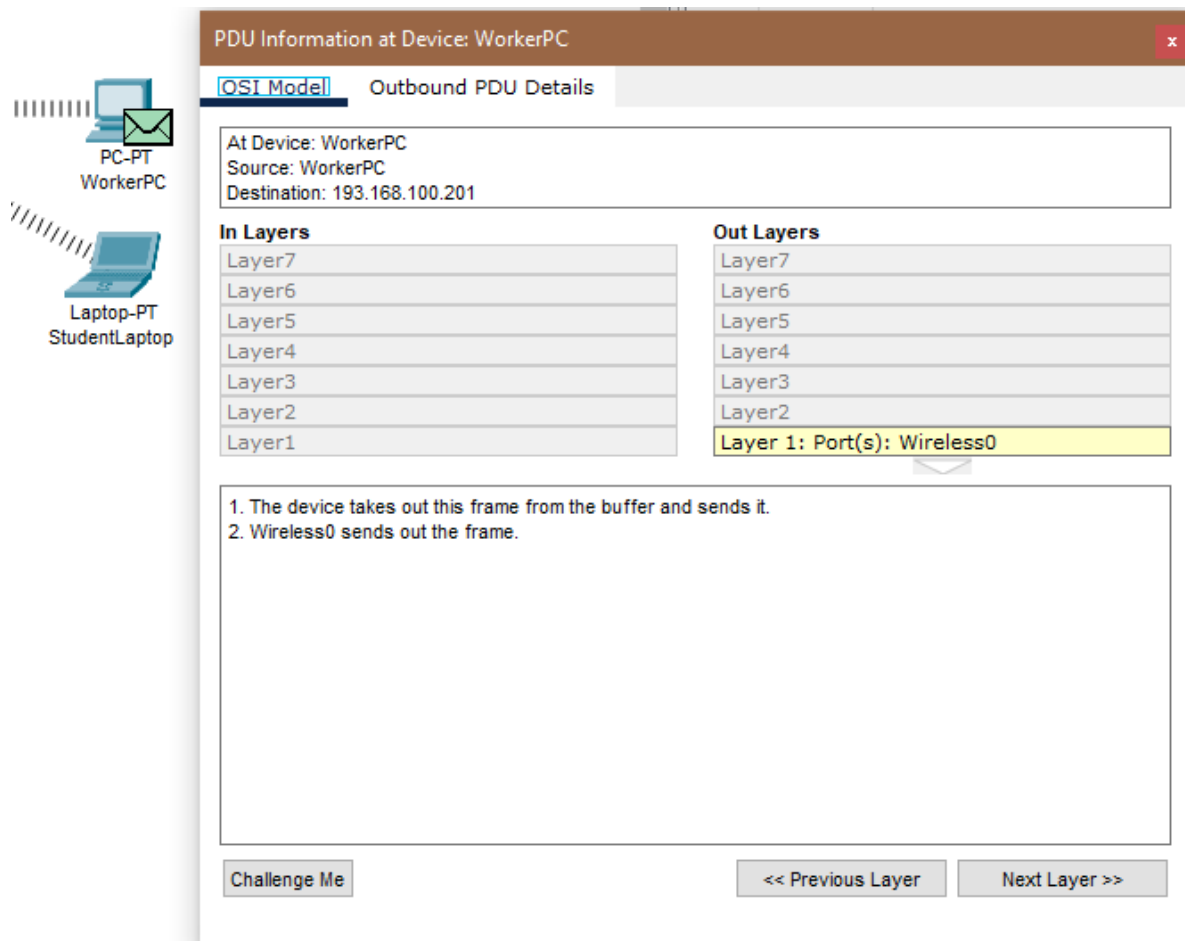
1. The HTTP client sends a HTTP request to the server.

Challenge Me

<< Previous Layer

Next Layer >>

The information is decoded and shown to the user though the Application layer (7) as a HTTP response, so the user can see it.



And everything was delivered correctly to the user. After this packet tracer seems like sends packages of confirmation and connection between the pcs and servers.

Bibliography

Sierra, C. A. (7 de Septiembre de 2024). *GitHub*. Obtained from Public Repository of Universidad Distrital Francisco José de Caldas: https://github.com/EngAndres/ud-public/blob/main/courses/computer-networking/workshops/Workshop_2.pdf